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Week 3 Circuits 2 Lab Experiment
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Experiment 1: MOSFET Characteristics:

Objective: The purpose of this Experiment is to understand and see in practice the forward bias effect and the voltage needed to reach proper current. And secondly, to understand that at a certain point adding voltage does not add more current.

Materials: 1 kOhlm Resistor, 4 2N7000 nmos resistors

3.1

Attempt	Vin [V]
1	1.713
2	1.756
3	1.793
4	1.692
5	1.732
Average	1.737

Attempt	Vin [V]	Vin [V]	Vin [V]
1	1.812	1.712	1.756
2	1.837	1.736	1.785
3	1.826	1.724	1.758
4	1.825	1.725	1.759
5	1.793	1.730	1.743
Average	1.819	1.725	1.760

Attempt	Vs [V]
1	0.109
2	0.116

3	1.101
4	0.113
5	0.111
Average	0.110

Vin [V]	Vs [V]	V_{TH} [V]
1.737	0.110	1.627

$$V_{TH} = V_{in} - V_s = 1.737 - 0.110 = 1.627V$$

3.2

Vin [V]	I_D [mA]
1.2	0.00735
1.3	0.03354
1.4	0.14706
1.5	0.576
1.6	1.873
1.7	5.032
1.79	10.011
1.8	11.164
1.81	12.021
1.9	18.154
2	19.310
2.1	19.356
2.2	19.364

V_{OV}	I_D [mA]	Gm [mA/V]
0.173	11.164	129.064

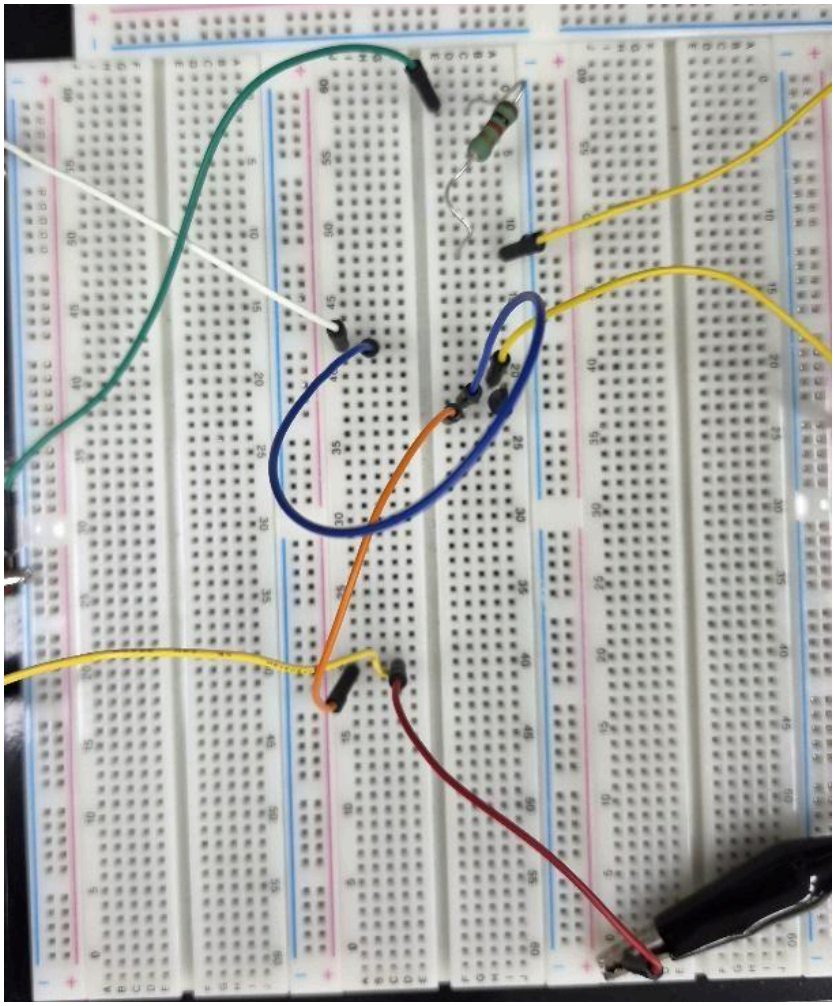
$V_{OV} = V_{GS} - V_{TH} = 1.8 - 1.627 = 0.173V$

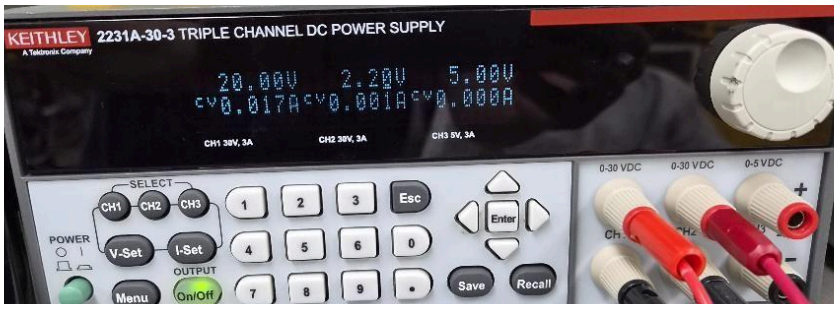
V_{GS} [V]	I_D [mA]	Gm [mA/V]
1.79	10.011	100.5
1.81	12.021	

$$g_m = \frac{\Delta I_D}{\Delta V_{GS}} = \frac{12.021 - 10.011}{1.81 - 1.79} = 100.5 mA/V$$

3.3

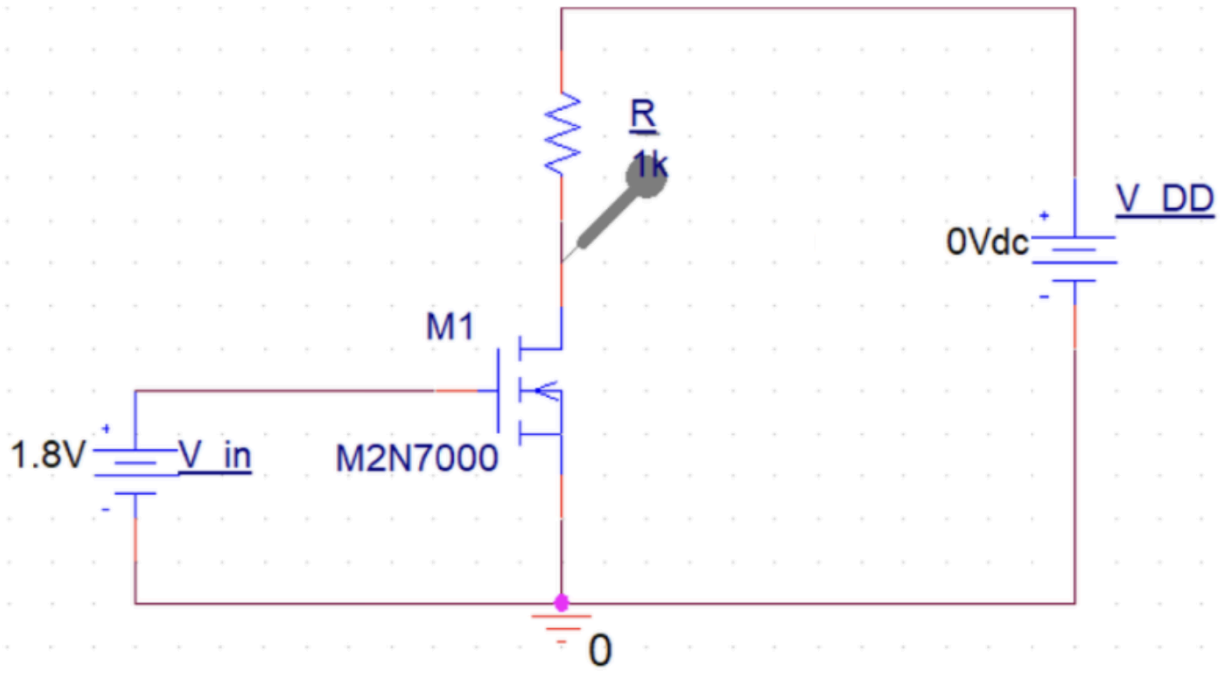
V_{DD} [V]	V_{DS} [V]	I_D [mA]
0	0.000	0.000667
1	0.022	0.869
2	0.057	1.734
3	0.134	2.553
4	0.528	3.113
5	1.354	3.229
6	2.231	3.353
7	3.125	3.466
8	4.002	3.573
9	4.857	3.743
10	5.719	3.833
11	6.583	3.971
12	7.412	4.121

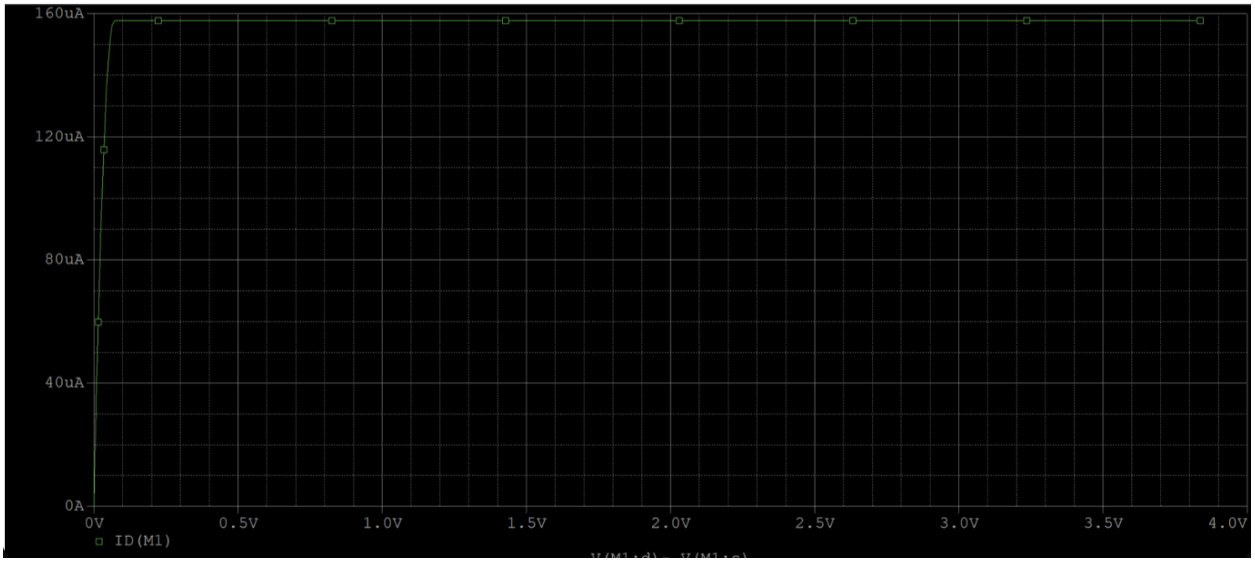


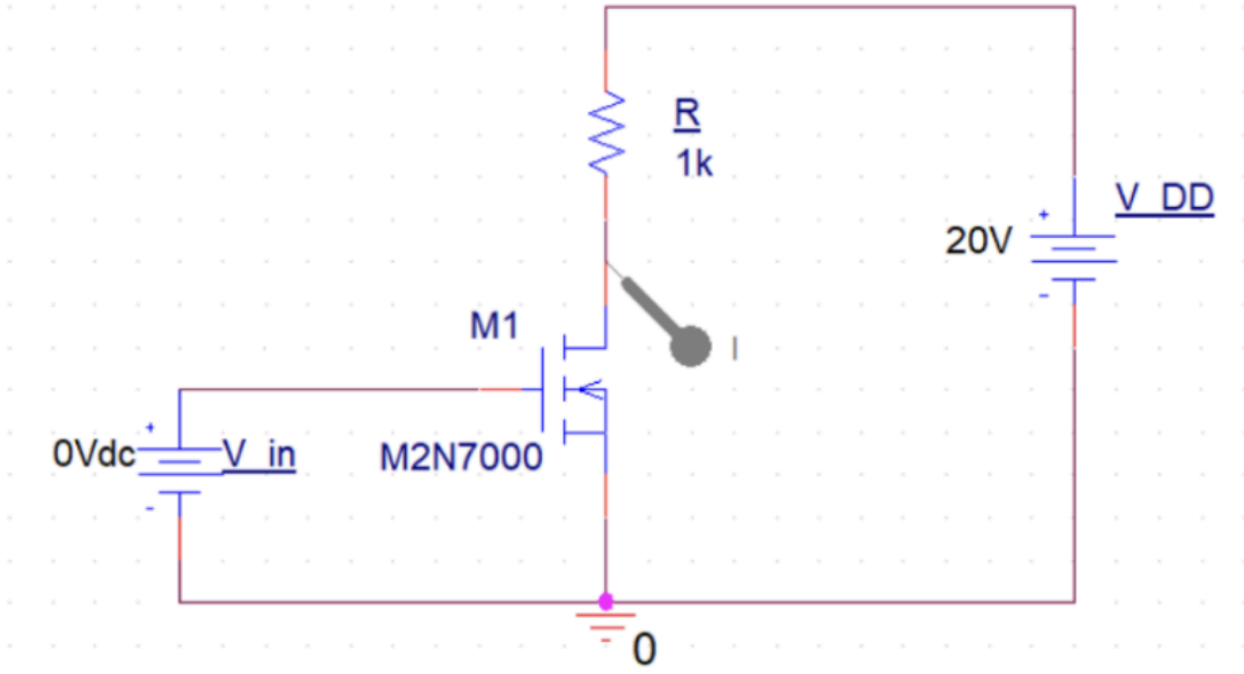

KEITHLEY
A Tektronix Company
2231A-30-3 TRIPLE CHANNEL DC POWER SUPPLY
20.00V 2.20V 5.00V
CV 0.017A CV 0.001A CV 0.000A
CH1 30V, 3A
CH2 30V, 3A
CH3 5V, 3A
SELECT
CH1
CH2
CH3
POWER
V-Set
I-Set
OUTPUT
On/Off
Menu
1
2
3
Esc
4
5
6
0
7
8
9
.
Enter
Save
Recall
0-30 VDC
0-30 VDC
0-5 VDC
+
-

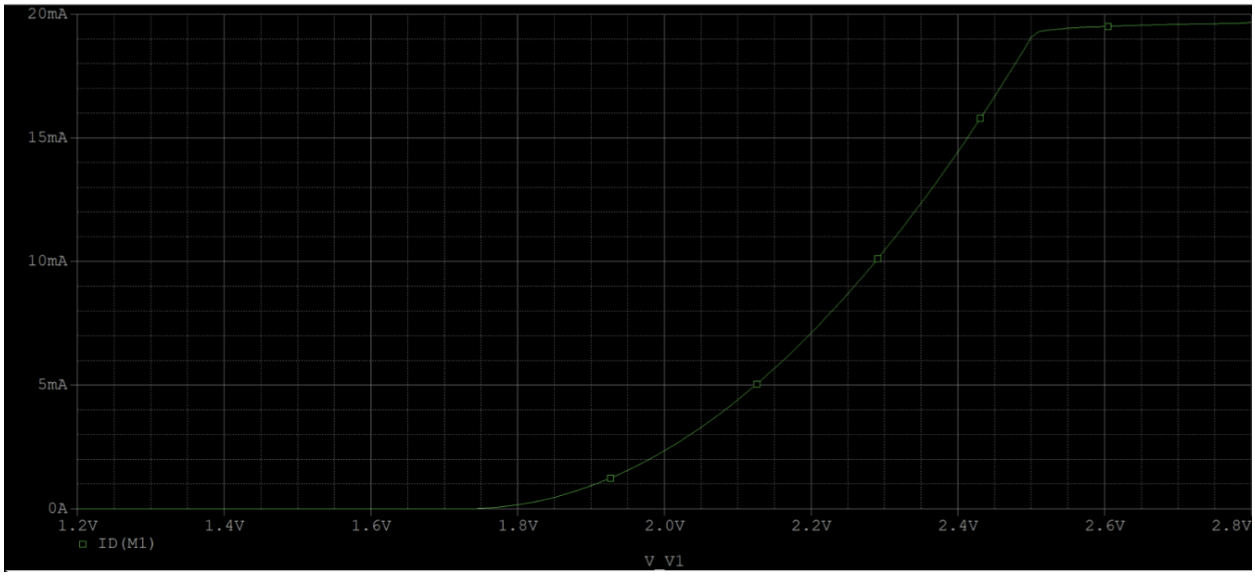

GW INSTEK GDM-8341
50000 Counts Dual Measurement Multimeter
DC
AUTO
M
19.364
50mA
INPUT
CAT II 600V
MAX 0.5A
FUSE T0.5A 250V
POWER
DC + AC
DCI
ACI
DCV
ACV
dB
dBm
Hz/P
LOCAL
2ND
RANGE +
Auto Enter
RANGE −
REL#
MATH
COMP
INT/EXT
REL
MX/MN
HOLD
TRIG
MENU
SHIFT EXIT

PSPICE


R
1k
V_DD
0Vdc
M1
1.8V
V_in
M2N7000
0


160uA
120uA
80uA
40uA
0A
0V
0.5V
1.0V
1.5V
2.0V
2.5V
3.0V
3.5V
4.0V
ID(M1)


R
1k
V_DD
20V
M1
0Vdc
V_in
M2N7000
0


20mA
15mA
10mA
5mA
0A
1.2V
1.4V
1.6V
1.8V
2.0V
2.2V
2.4V
2.6V
2.8V
ID(M1)
V_V1